

Richard B. Anderson

+1 (915) 801-6068 | richard.b.anderson@icloud.com | mrandrson.github.io/richard-react | github.com/mrandrson

RESEARCH INTERESTS

I work in computational astrophysics with an emphasis on planet formation, gravitational dynamics, and simulation-driven modeling. My current research studies the capture of planetesimals during protostellar collapse by combining analytic estimates with REBOUND N-body simulations and large parameter sweeps on HPC systems. More broadly, I am interested in disk evolution, turbulence, radiative and hydrodynamic processes, and other computationally intensive problems in astrophysics.

EDUCATION

The University of Texas at El Paso, El Paso, TX *Jan 2023 – May 2026 (expected)*
B.S. in Physics; B.S. in Mathematics
• Overall GPA: 3.82/4.00

RESEARCH EXPERIENCE

Undergraduate Research Assistant *El Paso, TX*
UTEP Astrophysics Theory Group
(Advisor: Dr. Kedron Silsbee) *Aug 2022 – Present*
• Develop and run REBOUND N-body simulations to study planetesimal capture during protostellar collapse.
• Derive analytic and numerical models for scattering, dissipation, and capture efficiency across encounter parameter space.
• Execute large simulation campaigns on UTEP's SLURM cluster and analyze outputs with Python-based post-processing workflows.

SULI Intern *Upton, NY*
Brookhaven National Laboratory
Nonproliferation and National Security Department *Jun 2025 – Aug 2025*
• Built discrete-element simulations in LIGGGHTS to study packing and geometry in pebble-bed reactor configurations.
• Performed OpenMC neutron and gamma-ray transport calculations across burnup stages to characterize spectral evolution.
• Automated parameter sweeps, data reduction, and feature extraction in Python for machine-learning-based burnup classification.
• Completed professional training in nonproliferation and international nuclear safeguards through the SULI program.

Undergraduate Research Intern *Arlington, TX*
UT Arlington
Neutrino and Rare Event Searches (NuRES) *May 2024 – Dec 2024*
• Developed waveform-analysis pipelines for NEXT detector data to extract decay constants, peak widths, and related pulse-shape observables.
• Simulated alpha-induced backgrounds in Geant4/Nexus to study ionization topologies in high-pressure xenon.
• Used nuDoBE to evaluate neutrinoless double-beta decay rates across effective-field-theory Wilson-coefficient parameter space.

Undergraduate Research Assistant, Missile Systems Innovation Team *El Paso, TX*
UTEP Aerospace Center *Sep 2023 – May 2024*
• Integrated drone hardware with Python/C++ control and tracking software within the ArduPilot ecosystem.
• Modeled ballistic trajectories and optimized launch parameters using numerical methods and simulation.

- Worked with interdisciplinary engineering teams using digital systems-engineering workflows to support design and testing.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant, Electromagnetism Laboratory

El Paso, TX

The University of Texas at El Paso, Department of Physics

Aug 2025 – Present

- Assist students with laboratory setup, data collection, and analysis in undergraduate electromagnetism labs.
- Provide real-time troubleshooting and conceptual support during experiments and post-lab work.

Academic Tutor

El Paso, TX

El Paso Community College, Student Success

Jul 2022 – Aug 2023

- Tutored college mathematics from remedial algebra through differential equations in one-on-one and drop-in settings.
- Supported both in-person and online sessions, adapting explanations to students with varied preparation levels.
- Coordinated with tutoring staff during high-demand periods and maintained detailed usage records.

PUBLICATIONS & MANUSCRIPTS

- Anderson, Richard B., and Kedron Silsbee. **Capture of Interstellar Planetesimals**. Manuscript in preparation.

CONFERENCE ABSTRACTS & PRESENTATIONS

- Anderson, Richard B., and Kedron Silsbee. **Capture of Freely-Floating Planetesimal Into Protoplanetary Disks**. APS Texas Section Fall Meeting (TAOF25), Abstract FB01.63, 2025. <https://meetings.aps.org/Meeting/TAOF25/Session/FB01.63>

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, Fortran, Mathematica, Bash

Scientific Software: REBOUND, OpenMC, Geant4/Nexus, LIGGGHTS, STAR-CCM+

Tools and Platforms: Git, Linux/Unix, SLURM, LaTeX, Markdown

AWARDS & HONORS

Outstanding Senior in Physics, *UTEP Department of Physics* 2026

National Science Foundation Graduate Research Fellowship Honorable Mention, *National Science Foundation* 2026

Terry Foundation Scholarship, *Terry Foundation* 2023–2027

Dean's List, *The University of Texas at El Paso* 2023–2025

COURI SURPASS Fellowship, *The University of Texas at El Paso* 2023

NuPEERS Nuclear Physics Summer School, *NuPEERS* 2023

REFERENCES

Dr. Kedron Silsbee | Assistant Professor

Department of Physics

The University of Texas at El Paso

ksilsbee@utep.edu